Computer Architecture – Chapter-wise MCQs and Numericals with Answers

Based on: Computer Architecture: A Quantitative Approach (Hennessy & Patterson, 6th Ed.)

Chapter 1: Performance and Quantitative Principles

Numerical Problems with Answers

1. IC = 8×10■, CPI = 1.5, Clock = 2 GHz. Find CPU time.
Answer: (8e8×1.5)/(2e9) = 0.6 s

2. 30% of program improved by 4×. Find speedup.
Answer: 1/(0.7 + 0.3/4) = 1/0.775 ≈ 1.29

3. System A takes 25s, B takes 15s. Speedup of B?
Answer: 25/15 = 1.67×

4. If voltage reduced by 20%, power changes by?
Answer: $(0.8^2) = 0.64 \rightarrow$ 36% reduction

5. If CPI=1.2 and IC=5e8 at 2.5GHz, find time.
Answer: (5e8×1.2)/2.5e9 = 0.24 s

Chapter 2: Memory Hierarchy

Numerical Problems with Answers

1. Hit=1ns, Miss rate=4%, Miss penalty=80ns. AMAT?
Answer: 1 + 0.04×80 = 4.2 ns

2. Cache=64KB, block=64B, 2-way. Sets?
Answer: 65536/(64×2)=512 sets

3. Miss rate=3%, accesses/instr=1.3. MPI?
Answer: 0.039

4. Page fault time=8ms, rate=1e-5. Overhead?
Answer: 8e-3 × 1e-5 = 8e-8 s

5. TLB hit=98%, miss penalty=100ns, hit=1ns. AMAT?
Answer: 1 + 0.02×100 = 3 ns

Chapter 3: Instruction-Level Parallelism

Numerical Problems with Answers

1. Ideal CPI=1, branch freq=20%, penalty=4 cycles, predict=90%. CPI?
Answer: 1 + 0.2×0.1×4 = 1.08

2. Pipeline depth=10, branch penalty=8 cycles, freq=15%. CPI?
Answer: 1 + 0.15×8 = 2.2

3. Issue width=4, achieved IPC=2.5. Utilization?
Answer: 2.5/4 = 62.5%

4. Base CPI=1, cache miss CPI=0.3, branch CPI=0.2. Total?
Answer: 1.5

5. If 25% instr depend, stall 2 cycles, CPI increase?
Answer: 0.25×2 = 0.5

Chapter 4: Data-Level Parallelism and GPUs

Numerical Problems with Answers

1. GPU = 3 TFLOPS, BW = 300 GB/s. Balance OI?
Answer: 3000/300 = 10 FLOP/byte

2. Kernel uses 256 threads/block, 8 blocks/SM. Threads/SM?
Answer: 2048

3. If divergence causes 30% idle, efficiency?
Answer: 70%

4. Each thread loads 8 bytes, does 32 FLOPs. OI?
Answer: 4 FLOP/byte

5. Memory latency=400 cycles, warps=40. Is latency hidden?
Answer: Yes (many warps to cover latency)

Chapter 5: Thread-Level Parallelism

Numerical Problems with Answers

1. 16 cores, serial fraction 5%. Speedup?
Answer: 1/(0.05+0.95/16)=10.3

2. MTTF=500h, MTTR=5h. Availability?
Answer: 500/505 ≈ 0.99

3. 40 threads, only 32 active. Utilization?
Answer: 80%

4. Cache line=64B, var=4B, false sharing between 2 threads. How many vars per line?
Answer: 16

5. Sync overhead=10%, new speedup with ideal 8×?
Answer: 8×0.9 = 7.2

Chapter 6: Warehouse-Scale Computers

Numerical Problems with Answers

1. Server uses 300W, IT load=10kW, total=15kW. PUE?
Answer: 15/10 = 1.5

2. Each request=5ms CPU, 2000 req/s. Cores needed?
Answer: 2000×0.005 = 10 CPU-seconds → 10 cores

3. Failure rate=2/day, replicas=3. Reliability improves by?
Answer: Very high; failure only if all 3 fail

4. Network BW=10Gbps, utilization=70%. Effective BW?
Answer: 7 Gbps

5. Storage replication=3×, raw=300TB. Usable?
Answer: 100 TB

Chapter 7: Domain-Specific Architectures

Numerical Problems with Answers

1. CPU=100 GFLOPS @100W, DSA=500 GFLOPS @50W. Perf/W gain?
Answer: (10 vs 2) $\rightarrow$ 5× better

2. Precision reduced halves data, bandwidth doubles. Speedup?
Answer: ≈2×

3. Matrix 1024×1024 multiply ops?
Answer: $2\times1024^3 \approx 2.1e9$ ops

4. TPU runs at 700 MHz, array 256×256 MACs. Ops/s?
Answer: $256^2\times7e8 \approx 4.6e13$ ops

5. If accelerator offloads 60% work at 10× speed, speedup?
Answer: 1/(0.4+0.6/10)=2.17